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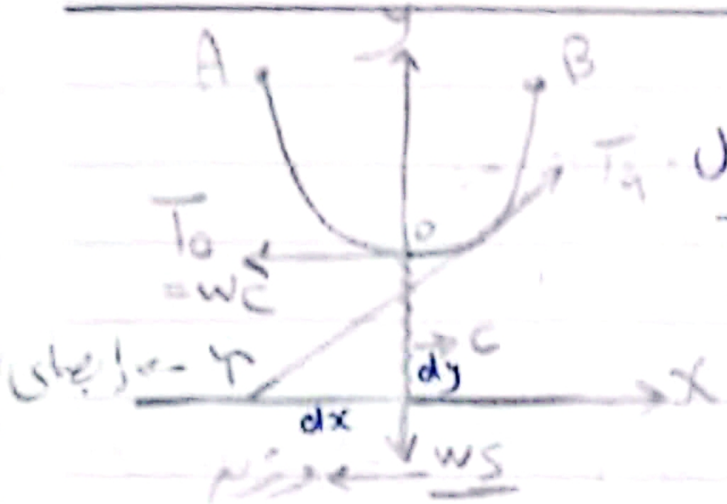
Subject :

Date :

السلاسل والخيوط :



ws ← وزن وحدة الأطوال

$T\cos\psi = T_0 = wc$

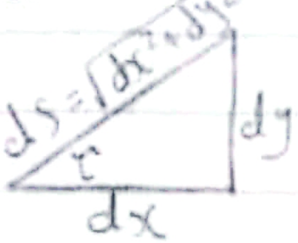
$T\sin\psi = ws$

$\tan\psi = \frac{s}{c}$

$s = c\tan\psi$

المعادلة الذاتية

$$\frac{dy}{dx} = \tan\psi = \frac{s}{c} \longrightarrow \frac{dy}{d\psi} = \frac{dy}{ds}\cdot\frac{ds}{d\psi} =$$



$s = c\tan\psi$ ← مشتقة

$ds = c\cdot\sec^2\psi\, d\psi$

$$\frac{dy}{d\psi} = \frac{dy}{ds}\cdot\frac{ds}{d\psi} = c\cdot\sin\psi\,\underline{\sec^2\psi}$$

$\frac{1}{\cos}$ sec

$$\frac{dy}{d\psi} = c\cdot\tan\psi\sec\psi$$

$dy = c\cdot\tan\psi\sec\psi\, d\psi$

تكامل $y = c\sec\psi + c_1$

at $y = c,\ \psi = 0$

$c = c + c_1 \Rightarrow c_1 = 0$

إذاً

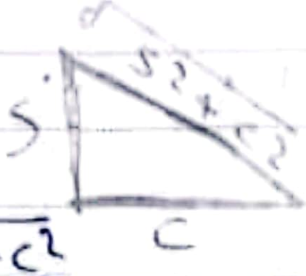
$\therefore$

$y = c\sec\psi$

$s = c\tan\psi$

$y = c\sec T$

$s = c\tan T$



$\sec T = \frac{1}{\cos T} = \frac{\sqrt{s^2+c^2}}{c}$

$y = \cancel{c} \cdot \frac{\sqrt{s^2+c^2}}{\cancel{c}}$

$y = \sqrt{s^2+c^2}$

$\frac{dx}{dT} = \frac{dx}{ds} \cdot \frac{ds}{dT} = c\cos T\sec^2 T =$

$= c\sec T$

$dx = c\sec T\, dT$

$x = c\ln(\sec T + \tan T) + C_2$

$x = 0\ ,\ T = 0$

$C_2 = 0$

$x = c\ln(\sec T + \tan T)$

$x = c\ln\left(\frac{s}{c} + \sqrt{1+\left(\frac{s}{c}\right)^2}\right.$

$x = c\ln(\sec T + \sqrt{\sec^2 T - 1})$ بالتعويض x ، y

$x = c\ln\left(\frac{y}{c} + \sqrt{\frac{y^2}{c^2} - 1}\right)$

$\frac{dy}{dx} = \tan T = \sqrt{\sec^2 T + 1} = \sqrt{\frac{y^2}{c^2} - 1}$

$\frac{dy}{dx} = \frac{\sqrt{y^2 - c^2}}{c}$

$\frac{dy}{\sqrt{y - c^2}} = \frac{1}{c}dx \qquad \cosh^{-1}\frac{y}{c} = \frac{x}{c} + C_3$

[illegible]

$$\cosh^{-1}\frac{y}{c} = \frac{x}{c} + \boxed{C_3} = 0$$

$$\frac{y}{c} = \cosh\frac{x}{c}$$

$$x = 0 \, , \, y = c$$

(باخذ cosh للطرفين)

$$y = c\cosh\frac{x}{c}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

تعريف

$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

$$\sinh x = \frac{e^{x} - e^{-x}}{2}$$

$$\frac{ds}{dx} = \sec T = \sqrt{1 + \tan^2 T}$$

اثبت

$$= \sqrt{1 + \frac{s^2}{c^2}} = \frac{\sqrt{c^2 + s^2}}{c}$$

$$\frac{ds}{\sqrt{s^2 + c^2}} = \frac{dx}{c}$$

$$T \sin T = ws$$

$$T \cos T = wc$$

$$T^2 = w^2 (c^2 + s^2)$$

تكامله

$$\sinh^{-1}\frac{s}{c} = \frac{x}{c} + C_4$$

$$S = c\sinh\frac{x}{c}$$